AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM129

Technologies in Practice

Final Examination
Spring Semester 2024/2025

Date:31/May/2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 3 h

Instructions:

- Total Marks: 50
- Total Number of Questions:
  - Question 1: 10
  - Question 2: 4
  - Question 3: 2
  - Question 4: 2
  - Question 5: 4
- Marks Distribution:
  - Question 1: 10
  - Question 2: 10
  - Question 3: 10
  - Question 4: 10
  - Question 5: 10
- Write your answer in the answer booklets. Answers given on the exam paper not be marked.
- The use of electronic devices that could have a memory is NOT permitted.
- Be sure you write your name and ID on the External Answer booklet.

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Question 1: Answer the following questions [10 Marks]

1. In ------- scheduling algorithm, the process that requests the CPU first is allocate the CPU first.
   a. RR
   b. FCFS
   c. SJF
   d. Priority.
2. Which of the following is NOT one of the four necessary conditions for a deadlo to occur?
   a. Mutual Exclusion
   b. Hold and Wait
   c. No Preemption
   d. Priority Scheduling
3. Which of the following is NOT a common media type for physical connections in lo networks?
   a. Copper
   b. Fiber optics
   c. Wireless
   d. Plastic
4. What is the main difference between open-loop and closed-loop control systems?
   a. Open-loop systems use feedback, while closed-loop systems do not.
   b. Closed-loop systems use feedback, while open-loop systems do not.
   c. Open-loop systems are more complex than closed-loop systems.
   d. Closed-loop systems cannot adjust outputs dynamically.
5. Which of the following is NOT a condition that will stop a robot?
   a. The program/driven action is finished
   b. There is an alarm condition
   c. The robot detects a human nearby
   d. There is a mechanical failure
6. What is the purpose of a light curtain in robotic safety systems?
   a. To detect pressure applied by a person stepping on it
   b. To provide illumination for operators in low-light areas
   c. To prevent automatic operation when someone enters the danger zone
   d. To serve as a physical barrier to block access to the robot
7. Which of the following guarding materials is commonly used in industrial settings to protec workers from robotic hazards?
   a. Plastic sheets
   b. Expanded metal guarding
   c. Wooden panels
   d. Glass barriers
8. Which of the following is NOT a common application of robotics?
   a. Industrial automation

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b. Medical surgery
c. Space exploration
d. Social media

9. What is the term used to describe a robot's ability to understand and interact with its environment?
   a. Artificial intelligence
   b. Machine learning
   c. Computer vision
   d. Sensing

10. Which of the following is NOT a function of an operating system?
   a. Managing memory and storage
   b. Providing a user interface
   c. Running applications
   d. Sending emails

Question 2: Answer the following questions [10 Marks]

a) Define: Operating system, Process, Process state, PCB? (2 Marks)
b) Explain with diagram the process states? (2 Marks)
c) Find the average waiting time (A.W.T) and average turnaround time (A.T.A.T) for executing the following process using: -
   1) Preemptive short job first (3 Marks)
   2) Non-preemptive short job first? (3 Marks)

Process	P1	P2	P3	P4	P5
Burst time	5	13	8	4	10
Arrival time	2	3	0	5	1

Question 3: Answer the following questions [10 Marks]

a) Compare and contrast the roles of sensors and actuators in an IoT system. How do they work together to enable automation?(4 Marks)
b) Suppose we have two resources, A, and B. A has 6 instances and B has 3 instances. Can the system execute the following processes without deadlock occurring? (6 Marks).

Process	Allocate		Maximum need	
	A	B	A	B
P1	1	1	2	2
P2	1	0	4	2
P3	1	0	3	2
P4	0	1	1	1

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P5	2	1	6	3

Question 4: Answer the following questions [10 Marks]

a) Match the following components of the Cisco IoT System with their roles.(4 Marks)

1. Network Connectivity	a) Enables communication between machines, such as temperature and oil sensors communicating with an onboard computer in cars.
2. Fog Computing	b) Provides scalable cybersecurity solutions, including operational technology (OT) security and IoT network security.
3. Data Analytics	c) Processes data closer to the network edge, reducing the burden on networks and enhancing real-time decision-making.
4. Security	d) Uses tools like Apache Hadoop and Kafka to analyze vast amounts of IoT-generated data for insights and trends.

b) Describe the three main zones around a robot and explain the safety measures required for each zone. (3 Marks)

c) Compare step mode and continuous mode in robot program testing. When should each be used, and what precautions are necessary?(3 Marks)

Question 5: Answer the following questions [10 Marks]

a) Match the following IoT protocols with their characteristics.(4 Marks)

1. MQTT	a) A lightweight messaging protocol with minimal overhead, providing high data integrity and security for remote environments.
2. CoAP	b) A type of API designed to make it easier for programs to interact over the Internet, often used in web-based IoT systems.
3. REST	c) A constrained application protocol designed for low-power and low-bandwidth IoT devices, enabling communication in sub-optimal conditions.
4. HTTP	d) A widely used protocol for transferring data over the web, but less efficient for IoT due to higher overhead compared to other protocol

b) Rank the following IoT communication protocols from lowest to highest in terms of power consumption:(2 Marks)

1) Zigbee
2) Wi-Fi
3) Bluetooth Low Energy (BLE)
4) Cellular (3G/4G/5G)

c) You want to build a smart home system where you can control lights, temperature, and security using a mobile app. Which IoT component would enable communication between the mobile app and the smart devices?(2 Marks).

d) What are the main challenges or considerations in terms of security in IoT deployments? (2 Marks)

----------End questions

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